

Hotel Booking Analysis

- Poorvi More

Project Report

1. Project Overview

Objective

Managing hotel bookings through multiple Excel files can quickly become difficult as the business grows. Important information is scattered across different workbooks, making it time-consuming to track occupancy, identify booking trends, or understand guest behavior.

The goal of this project was to transform those raw booking records into meaningful business insights. Using Python for data preparation, SQL for business analysis, and Power BI for visualization, I developed an interactive dashboard that helps hotel management monitor performance, understand booking patterns, and make informed decisions based on data instead of assumptions.

2. Business Problem

The hotel maintained its booking history in separate Excel files, making it challenging to answer everyday business questions efficiently.

Management wanted answers to questions such as:

- Which rooms are occupied most frequently?
- Which months generate the highest occupancy?
- Who are the hotel's repeat guests?
- Are weekends busier than weekdays?
- When should room maintenance be scheduled?
- How can occupancy be improved during slower seasons?

Without a centralized reporting system, these insights required manual effort and were often unavailable when decisions needed to be made.

3. Dataset summary

Data Sources

- SHELAR AGRO BOOKING FINAL.xlsx
- Shelar Booking_2024.xlsx

Time Period

July 2021 – June 2026

Hotel Capacity

6 Rooms

Information Available

The dataset contains daily booking information, including:

- Booking Date
- Room Number
- Guest Name
- Occupancy Status
- Year and Month
- Weekday
- Weekend Indicator

This historical data forms the foundation for analyzing occupancy trends, guest behavior, and room utilization.

4. Data Preparation

Before analysis could begin, the booking records required significant cleaning and restructuring.

Using Python and the Pandas library, I combined data from multiple Excel files into a single structured dataset. During this process, inconsistent formats were standardized, duplicate records were removed, guest names were cleaned, and additional date-related columns were created to simplify future analysis.

The final cleaned dataset was exported as a CSV file, ready for SQL analysis and Power BI visualization.

	Date	Year	Month	Day	Weekday	Room_No	Guest_Name	Status
0	2021-07-08	2021	July	8	Thursday	1	<NA>	Vacant
1	2021-07-09	2021	July	9	Friday	1	<NA>	Vacant
2	2021-07-10	2021	July	10	Saturday	1	Mayuresh Kawale	Occupied
3	2021-07-11	2021	July	11	Sunday	1	Mayuresh Kawale	Occupied
4	2021-07-12	2021	July	12	Monday	1	<NA>	Vacant

Rows: 4728

Unique Rooms: [1 2 3 4 5 6]

	Date	Year	Month	Day	Weekday	Room_No	Guest_Name	Status
0	2024-01-01	2024	January	1	Monday	1	<NA>	Vacant
1	2024-01-01	2024	January	1	Monday	2	<NA>	Vacant
2	2024-01-01	2024	January	1	Monday	3	<NA>	Vacant
3	2024-01-01	2024	January	1	Monday	4	<NA>	Vacant
4	2024-01-01	2024	January	1	Monday	5	<NA>	Vacant
5	2024-01-01	2024	January	1	Monday	6	<NA>	Vacant
6	2024-01-02	2024	January	2	Tuesday	1	Mayuri Banne	Occupied
7	2024-01-02	2024	January	2	Tuesday	2	Mayuri Banne	Occupied
8	2024-01-02	2024	January	2	Tuesday	3	<NA>	Vacant
9	2024-01-02	2024	January	2	Tuesday	4	<NA>	Vacant
10	2024-01-02	2024	January	2	Tuesday	5	<NA>	Vacant
11	2024-01-02	2024	January	2	Tuesday	6	<NA>	Vacant

Total Records: 5292

Unique Rooms: [1 2 3 4 5 6]

	Date	Year	Month	Day	Weekday	Room_No	Guest_Name	Status
0	2021-07-08	2021	July	8	Thursday	1	NaN	Vacant
1	2021-07-08	2021	July	8	Thursday	2	NaN	Vacant
2	2021-07-08	2021	July	8	Thursday	3	NaN	Vacant
3	2021-07-08	2021	July	8	Thursday	4	NaN	Vacant
4	2021-07-08	2021	July	8	Thursday	5	NaN	Vacant
10015	2026-06-30	2026	June	30	Tuesday	2	NaN	Vacant
10016	2026-06-30	2026	June	30	Tuesday	3	NaN	Vacant
10017	2026-06-30	2026	June	30	Tuesday	4	NaN	Vacant
10018	2026-06-30	2026	June	30	Tuesday	5	NaN	Vacant
10019	2026-06-30	2026	June	30	Tuesday	6	NaN	Vacant

(10020, 8)

5. Data Cleaning & Exploratory Analysis

To ensure reliable insights, the data was carefully validated and cleaned.

The preparation process included:

- Removing duplicate records

- Standardizing column names and date formats
- Cleaning guest names and formatting inconsistencies
- Removing invalid booking records
- Identifying occupied and vacant rooms
- Creating additional fields such as Year, Month, Day, Weekday, and Weekend

These steps improved data quality and ensured that all subsequent analyses were based on accurate information.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10020 entries, 0 to 10019
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Date        10020 non-null  object
1   Year        10020 non-null  int64
2   Month       10020 non-null  object
3   Day         10020 non-null  int64
4   Weekday     10020 non-null  object
5   Room_No     10020 non-null  int64
6   Guest_Name  2619 non-null   object
7   Status      10020 non-null  object
dtypes: int64(3), object(5)
memory usage: 626.4+ KB
None
```

	Date	Year	Month	Day	Weekday	Room_No	Guest_Name	Status
0	2021-07-08	2021	July	8	Thursday	1	NaN	Vacant
1	2021-07-08	2021	July	8	Thursday	2	NaN	Vacant
2	2021-07-08	2021	July	8	Thursday	3	NaN	Vacant
3	2021-07-08	2021	July	8	Thursday	4	NaN	Vacant
4	2021-07-08	2021	July	8	Thursday	5	NaN	Vacant

```
(10020, 8)

Index(['date', 'year', 'month', 'day', 'weekday', 'room_no', 'guest_name',
      'status'],
      dtype='object')
```

```
Missing Values
date          0
year          0
month         0
day           0
weekday       0
room_no       0
guest_name    7220
status        0
month_no      0
quarter       0
is_weekend    0
dtype: int64

Duplicate Rows
0

Room Numbers
[1 2 3 4 5 6]

Status
status
Vacant      7220
Occupied    2619
Name: count, dtype: int64

Date Range
2021-07-08 00:00:00
2026-06-30 00:00:00
```

6. Business Analysis Using SQL

SQL was used to answer key operational questions that hotel management encounters regularly.

The analysis focused on:

- Overall occupancy rate
- Monthly booking trends
- Most frequently booked rooms
- Loyal and repeat guests
- Weekend versus weekday occupancy
- Number of fully occupied days
- Average occupied rooms per day
- Unique guest count
- Guests returning across multiple years

- Room performance rankings
- Monthly booking rankings

These insights provide management with a clear understanding of hotel performance and customer behavior.

1. What is the hotel's occupancy rate?

	occupancy_rate
▶	25.72

2. Which months are busiest?

	year	month	occupied_rooms
▶	2021	July	15
	2021	August	19
	2021	September	15
	2021	October	40
	2021	November	91
	2021	December	104
	2022	January	67
	2022	February	64
	2022	March	27
	2022	April	40
	2022	May	110
	2022	June	24
	2022	July	10
	2022	August	17
	2022	October	54
	2022	November	38
	2022	December	74
	2023	March	16
	2023	April	42
	2023	May	83
	2023	June	37
	2023	July	29
	2023	August	27
	2023	September	17
	2023	October	19
	2023	November	61
	2023	December	77
	2024	January	53
	2024	February	26
	2024	March	19
	2024	April	25
	2024	May	109
	2024	June	67
	2024	July	20
	2024	August	13
	2024	September	3
	2024	October	22
	2024	November	39
	2024	December	71
	2025	January	71
	2025	February	16
	2025	March	2
	2025	April	50
	2025	May	71
	2025	June	45
	2025	July	14
	2025	August	21
	2025	September	14
	2025	October	55
	2025	November	45
	2025	December	76
	2026	January	46
	2026	February	24
	2026	March	22
	2026	April	50
	2026	May	110
	2026	June	84

3. Which room is booked most often?

	room_no	total_bookings
▶	1	695
	2	561
	3	423
	4	341
	5	276
	6	204

4. Who are the most loyal customers?

	guest_name	stays
▶	Prakash Bhatt	37
	Kishore Hajare	24
	Joby Cyrus	20
	Suresh V.	17
	Yatin Gharat	16
	Nikhil Patil	16
	Hitachi Group	15
	Bhavesb Modi	15
	Kulkarni	14
	Manish Patel	12

5. Are weekends busier?

1: Weekend 2: Weekday

	is_weekend	occupied_rooms
▶	1	1092
	0	1408

6. How often was the hotel full?

	date	occupied_rooms
►	2021-07-17 00:00:00	6
	2021-09-25 00:00:00	6
	2021-10-16 00:00:00	6
	2021-10-17 00:00:00	6
	2021-11-04 00:00:00	6
	2021-11-06 00:00:00	6
	2021-11-07 00:00:00	6
	2021-11-12 00:00:00	6
	2021-11-20 00:00:00	6
	2021-11-27 00:00:00	6
	2021-12-04 00:00:00	6
	2021-12-11 00:00:00	6
	2021-12-17 00:00:00	6
	2021-12-20 00:00:00	6
	2021-12-21 00:00:00	6
	2021-12-22 00:00:00	6
	2021-12-23 00:00:00	6
	2021-12-25 00:00:00	6
	2021-12-26 00:00:00	6
	2021-12-28 00:00:00	6
	2021-12-30 00:00:00	6
	2021-12-31 00:00:00	6
	2022-01-29 00:00:00	6
	2022-02-10 00:00:00	6

	2022-02-17 00:00:00	6
	2022-02-18 00:00:00	6
	2022-02-25 00:00:00	6
	2022-04-09 00:00:00	6
	2022-04-14 00:00:00	6
	2022-05-08 00:00:00	6
	2022-05-10 00:00:00	6
	2022-05-13 00:00:00	6
	2022-05-14 00:00:00	6
	2022-05-19 00:00:00	6
	2022-05-21 00:00:00	6
	2022-05-24 00:00:00	6
	2022-05-27 00:00:00	6
	2022-05-28 00:00:00	6
	2022-05-29 00:00:00	6
	2022-06-17 00:00:00	6
	2022-06-18 00:00:00	6
	2022-07-16 00:00:00	6
	2022-10-27 00:00:00	6
	2022-10-28 00:00:00	6
	2022-10-29 00:00:00	6
	2022-11-14 00:00:00	6
	2022-11-19 00:00:00	6

2022-12-23 00:00:00	6	2023-12-26 00:00:00	6
2022-12-24 00:00:00	6	2023-12-30 00:00:00	6
2022-12-29 00:00:00	6	2023-12-31 00:00:00	6
2022-12-31 00:00:00	6	2024-01-06 00:00:00	6
2023-04-29 00:00:00	6	2024-01-12 00:00:00	6
2023-05-09 00:00:00	6	2024-01-13 00:00:00	6
2023-05-10 00:00:00	6	2024-01-26 00:00:00	6
2023-05-27 00:00:00	6	2024-01-27 00:00:00	6
2023-05-28 00:00:00	6	2024-02-03 00:00:00	6
2023-05-31 00:00:00	6	2024-02-10 00:00:00	6
2023-06-08 00:00:00	6	2024-03-09 00:00:00	6
2023-07-08 00:00:00	6	2024-05-04 00:00:00	6
2023-07-14 00:00:00	6	2024-05-17 00:00:00	6
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2023-08-13 00:00:00	6	2024-05-28 00:00:00	6
2023-09-30 00:00:00	6	2024-05-31 00:00:00	6
2023-11-13 00:00:00	6	2024-06-05 00:00:00	6
2023-11-17 00:00:00	6	2024-06-08 00:00:00	6
2023-11-18 00:00:00	6	2024-06-15 00:00:00	6
2023-11-25 00:00:00	6	2024-06-16 00:00:00	6
2023-12-16 00:00:00	6	2024-11-02 00:00:00	6
2023-12-23 00:00:00	6	2024-11-03 00:00:00	6

2024-11-09 00:00:00	6	2025-10-04 00:00:00	6
2024-12-21 00:00:00	6	2025-10-22 00:00:00	6
2024-12-22 00:00:00	6	2025-10-24 00:00:00	6
2024-12-27 00:00:00	6	2025-10-25 00:00:00	6
2024-12-28 00:00:00	6	2025-11-08 00:00:00	6
2025-01-10 00:00:00	6	2025-12-06 00:00:00	6
2025-01-11 00:00:00	6	2025-12-22 00:00:00	6
2025-01-16 00:00:00	6	2025-12-25 00:00:00	6
2025-01-17 00:00:00	6	2025-12-26 00:00:00	6
2025-01-18 00:00:00	6	2025-12-27 00:00:00	6
2025-01-19 00:00:00	6	2025-12-28 00:00:00	6
2025-01-30 00:00:00	6	2025-12-31 00:00:00	6
2025-04-05 00:00:00	6	2026-01-03 00:00:00	6
2025-04-19 00:00:00	6	2026-01-09 00:00:00	6
2025-05-03 00:00:00	6	2026-01-10 00:00:00	6
2025-06-05 00:00:00	6	2026-01-24 00:00:00	6
2025-06-06 00:00:00	6	2026-01-25 00:00:00	6
2025-06-20 00:00:00	6	2026-02-14 00:00:00	6
2025-06-21 00:00:00	6	2026-04-30 00:00:00	6
2025-07-15 00:00:00	6	2026-05-02 00:00:00	6
2025-08-15 00:00:00	6	2026-05-12 00:00:00	6
2025-08-16 00:00:00	6	2026-05-13 00:00:00	6
2025-10-03 00:00:00	6	2026-05-23 00:00:00	6
2026-05-28 00:00:00	6		
2026-05-29 00:00:00	6		
2026-05-30 00:00:00	6		
2026-05-31 00:00:00	6		
2026-06-02 00:00:00	6		
2026-06-07 00:00:00	6		
2026-06-08 00:00:00	6		
2026-06-12 00:00:00	6		
2026-06-13 00:00:00	6		
2026-06-26 00:00:00	6		

7. Average Occupied Rooms Per Day

avg_rooms
▶ 3.25

8. Unique Guests

unique_guests
▶ 861

9. Guests Visiting Across Multiple Years

	guest_name	years_visited			
►	Nikhil Patil	4		Mr. Fahad	2
	Sachin Gajare	4		Nasir Shaikh	2
	Pranay Shinde	3		Nikhil	2
	Aniket Patil	3		Aniket	2
	Patil	3		Ninave	2
	Prakash Bhatt	3		Nitin Patil	2
	Pratik Patole	3		Parag	2
	Vaibhav Ambre	3		Aakash Bankar	2
	Vishal Shinde	3		Abhiraj Kashid	2
	Arvind Mhatre	2		Mayur	2
	Ashish Asangao	2		Prathamesh	2
	Chetan Wajekar	2		Aditya More	2
	Deepak Bagate	2		Ravindra Pise	2
	Dev Gajjar	2		Ajay Jadhav	2
	Dinesh Kadam	2		Sagar Mhatre	2
	Gautam	2		Salim	2
	Ioc Nishant	2		Sambhaji Yadav	2
	Jitu Koli	2		Sandeep More	2
	Kishor Tate	2		Ajay Kandpile	2
	Kishore Hajare	2		Shafi Shaikh	2
	Kulkarni	2		Sharad Majlekar	2
	Manoj Jadhav	2		Subhrajit	2
	Mehta	2		Sumedh Patil	2
	Mhatre	2			
	Vaibhav	2			
	Sanjay Patil	2			
	Vijay Patil	2			
	Vishal Daye	2			
	Amol	2			
	Vishal Wadkar	2			
	Vivek	2			
	Yatin Gharat	2			

10. Executive Summary

	total_room_nights	occupied_room_nights	vacant_room_nights	occupancy_rate	unique_guests
►	9720	2500	7220	25.72	861

11. Ranking Rooms

	room_no	bookings	room_rank
▶	1	695	1
	2	561	2
	3	423	3
	4	341	4
	5	276	5
	6	204	6

12. Monthly Booking Rank

	year	month	bookings	monthly_rank
▶	2021	December	104	1
	2021	November	91	2
	2021	October	40	3
	2021	August	19	4
	2021	July	15	5
	2021	September	15	5

2022	May	110	1
2022	December	74	2
2022	January	67	3
2022	February	64	4
2022	October	54	5
2022	April	40	6
2022	November	38	7
2022	March	27	8
2022	June	24	9
2022	August	17	10
2022	July	10	11

2023	May	83	1
2023	December	77	2
2023	November	61	3
2023	April	42	4
2023	June	37	5
2023	July	29	6
2023	August	27	7
2023	October	19	8
2023	September	17	9
2023	March	16	10

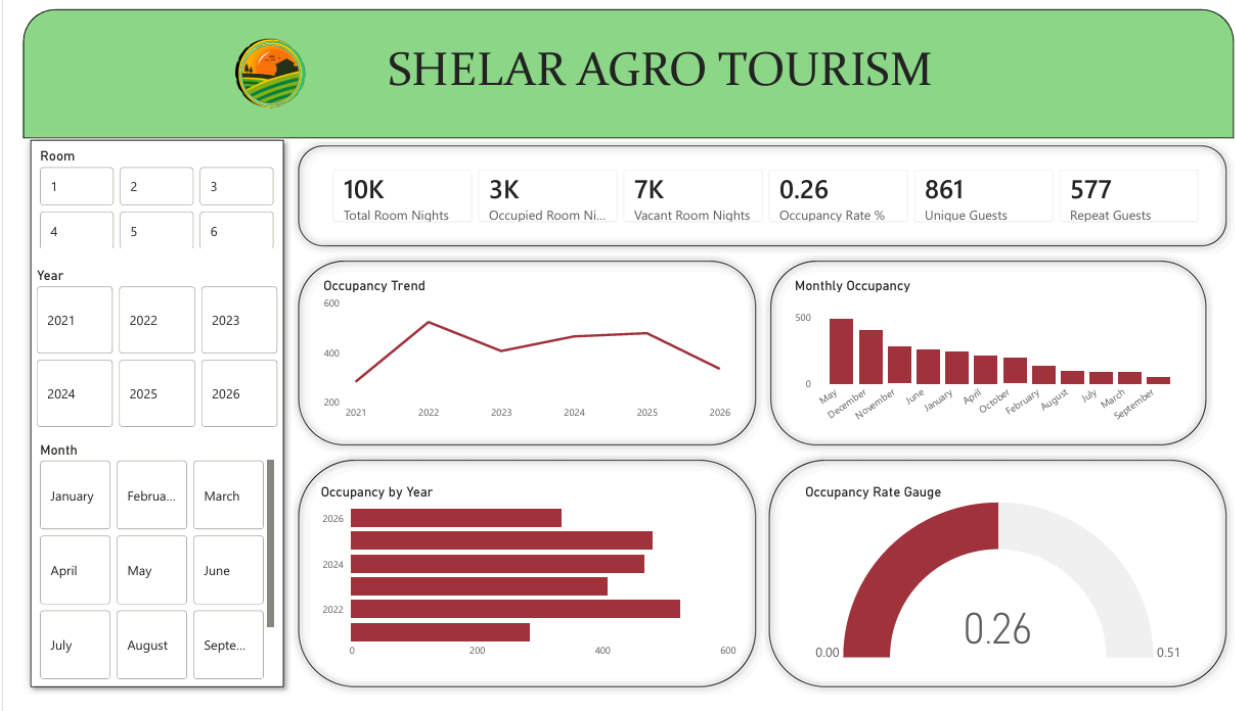
2024	May	109	1
2024	December	71	2
2024	June	67	3
2024	January	53	4
2024	November	39	5
2024	February	26	6
2024	April	25	7
2024	October	22	8
2024	July	20	9
2024	March	19	10
2024	August	13	11
2024	September	3	12
2025	December	76	1
2025	January	71	2
2025	May	71	2
2025	October	55	3
2025	April	50	4
2025	June	45	5
2025	November	45	5
2025	August	21	6
2025	February	16	7
2025	July	14	8
2025	September	14	8
2025	March	2	9
2026	May	110	1
2026	June	84	2
2026	April	50	3
2026	January	46	4
2026	February	24	5
2026	March	22	6

7. Dashboard in Power BI

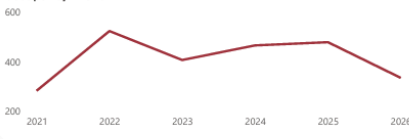
After completing the analysis, the results were presented in an interactive Power BI dashboard.

1. Executive Dashboard

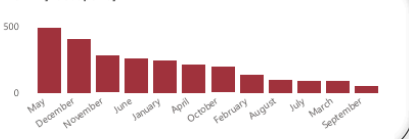
Provides an overview of key business metrics, including occupancy, bookings, and overall hotel performance.



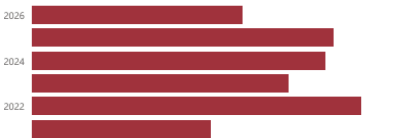
Occupancy Trend



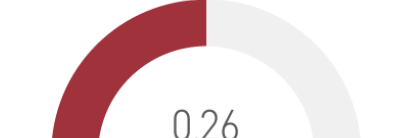
Monthly Occupancy



Occupancy by Year

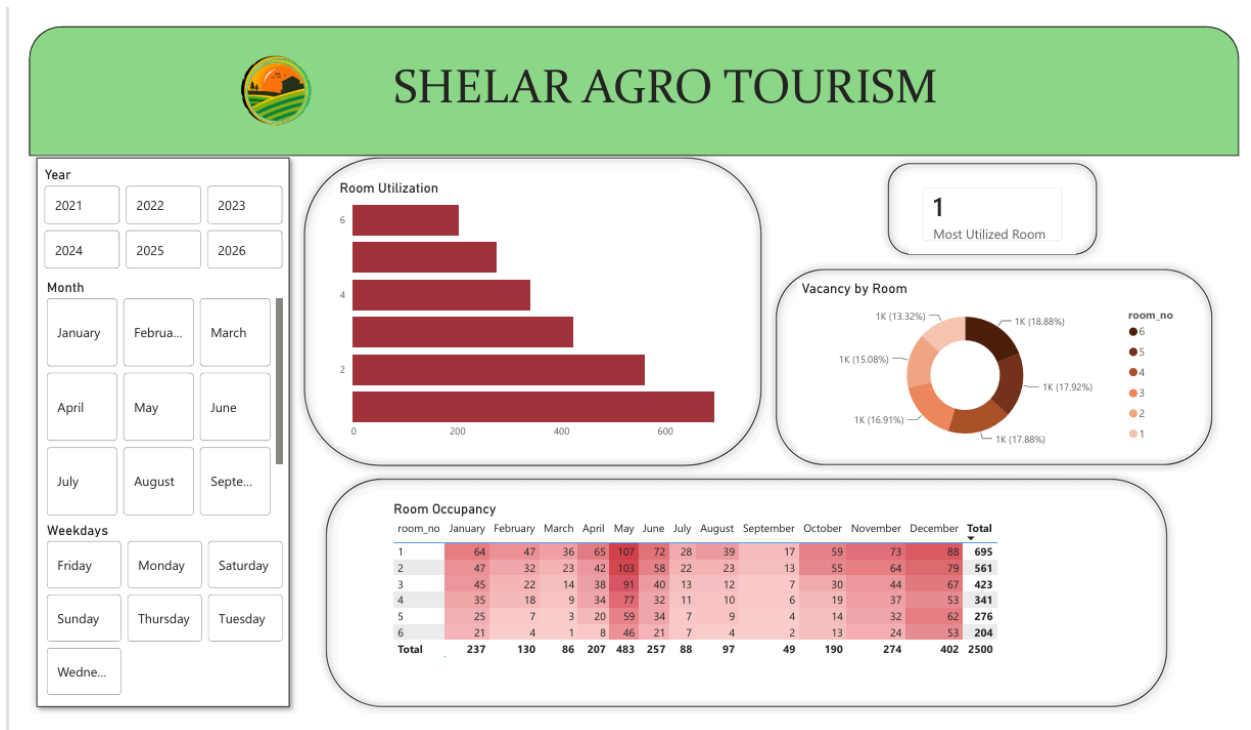


Occupancy Rate Gauge



2. Room Performance

Highlights the most and least utilized rooms, helping management optimize room allocation and maintenance schedules.



3. Guest Analysis

Identifies repeat visitors, unique guests, and long-term customer trends to support customer relationship strategies.



SHELAR AGRO TOURISM

861

Unique Guests

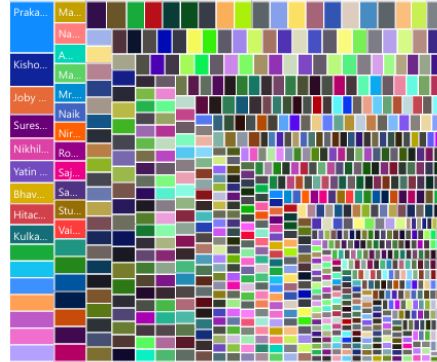
577

Repeat Guests

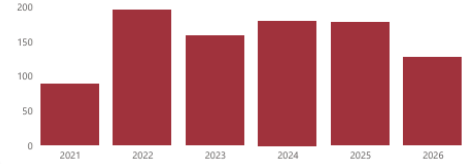
11.29

Average Stay

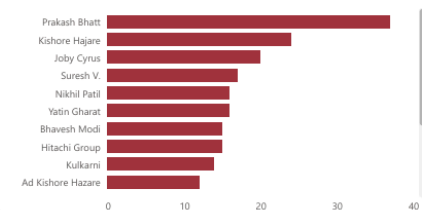
Guest Visit Frequency



Guests by Year



Top 10 Guest



4. Seasonal Trends

Displays booking patterns across months and years, helping management prepare for peak seasons and improve occupancy during quieter periods.



SHELAR AGRO TOURISM

Status

Occupied

Vacant

Room Number

1

2

3

4

5

6

month

January

Februa...

March

April

May

June

July

August

Septe...

Year

2021

2022

2023

2024

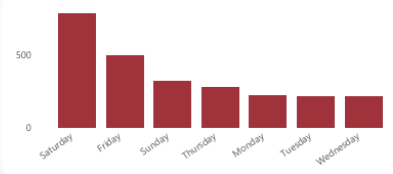
2025

2026

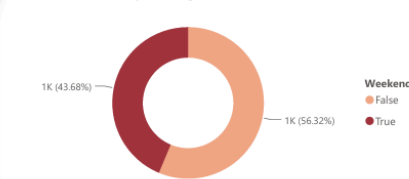
Monthly Occupancy Trend



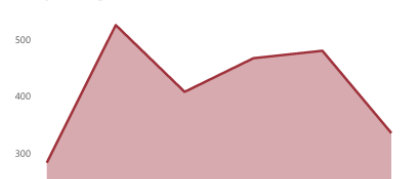
Weekday Analysis



Weekend vs Weekday Bookings



Yearly Booking Trend



Interactive filters allow users to explore the data by year, month, or room, making the dashboard useful for both strategic planning and day-to-day operations.

8. Business Recommendations

Based on the analysis, the following recommendations can help improve hotel operations and overall business performance:

1. Increase Occupancy During Off-Peak Months

The dashboard identifies months with lower booking activity. Management can introduce seasonal discounts, promotional packages, or weekend offers during these periods to attract more guests and improve occupancy.

2. Optimize Room Utilization

Rooms with consistently lower occupancy should be reviewed for pricing, amenities, or maintenance. Improving these rooms or adjusting their pricing strategy can help balance demand across all six rooms.

3. Reward Loyal Guests

The analysis identifies repeat visitors over multiple years. Introducing a loyalty program, personalized offers, or exclusive discounts can strengthen customer relationships and encourage repeat bookings.

4. Schedule Maintenance Strategically

Maintenance should be planned during periods of lower occupancy to minimize revenue loss and avoid disrupting guests during peak booking seasons.

5. Improve Weekend Revenue

If weekends consistently experience higher occupancy, management can introduce premium pricing or value-added packages to maximize revenue while maintaining guest satisfaction.

6. Use Seasonal Trends for Better Planning

Historical booking trends can help forecast future demand, allowing the hotel to prepare staffing, inventory, and marketing campaigns more effectively during peak and off-peak seasons.

7. Monitor Business Performance Regularly

The Power BI dashboard should be updated regularly with new booking data so management can monitor occupancy, room performance, and guest trends in real time, enabling faster and more informed decision-making.

9. Business Impact

Implementing these recommendations can provide several business benefits:

- Increase room occupancy and overall revenue.
- Improve customer retention through targeted loyalty initiatives.
- Optimize room allocation and maintenance scheduling.
- Reduce manual reporting effort with automated dashboards.
- Enable data-driven decision-making across hotel operations.
- Support long-term business planning using historical booking trends.

10. Conclusion

This project demonstrates a complete end-to-end data analytics workflow, starting from raw Excel booking records through data cleaning, SQL-based business analysis, and interactive Power BI dashboard development. The analysis provides valuable insights into occupancy trends, room utilization, guest behavior, and seasonal demand, enabling hotel management to make informed operational and strategic decisions. The project highlights practical skills in Python, SQL, Power BI, data transformation, business intelligence, and analytical storytelling.